

ClashArena — Kata (updated, with numbered references)

Online esports tournament platform — updated to number every item referenced in the architecture-characteristics deliverable

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1. About this kata

A kata is a short, high-level, domain-targeted description of an IT system, used as the starting point for architectural thinking. A kata is composed of four parts — Description, Users, Requirements and Additional context — from which the architect derives the system's architecture characteristics.

2. The kata

2.1 Description

ClashArena is an online platform that runs competitive video-game tournaments end to end for a set of supported titles — one individual 1v1 title and one 5v5 team title. Organisers create tournaments; players and teams register and check in; the platform seeds them and generates qualifiers, group stages and elimination brackets; matches are played on external game servers that report their results back; the platform validates each result, resolves disputes, advances winners, updates player and team ratings, and drives public leaderboards, tournament and match pages, and live-stream overlays in near real time.

The defining property of the domain is its load shape. Between events the platform is quiet, but during marquee events (majors, finals) the number of active competitors — and, far more sharply, the number of spectators — rises by orders of magnitude for a few hours before falling back to baseline.

2.2 Users

- U1. Players — individual competitors with profiles and ratings.
- U2. Teams and team captains — rosters, player roles, team registration.
- U3. Organisers / tournament admins — create and operate tournaments, seeding, scheduling.

- U4. Referees / moderators — verify results, resolve disputes, handle no-shows and forfeits.
- U5. Spectators / fans — the largest and most volatile group; read brackets, leaderboards and match pages and watch live streams.
- U6. Casters / broadcasters — consume live data for stream overlays.
- U7. Platform administrators — accounts, bans, anti-cheat and integrity.

Expected numbers (these make the two load characteristics concrete):

- U8. Registered accounts: expected to grow from about 50,000 to 500,000 over ~24 months.
- U9. Concurrent competitors in a typical weekly cup: a few thousand.
- U10. Spectators during a major final: short bursts of about 100,000–500,000 concurrent read/stream clients for a few hours, returning to a ~5,000 baseline afterwards.
- U11. Match results ingested: several thousand per hour during peak event windows.

2.3 Requirements

1. R1. Account registration and authentication; player profiles; team creation with rosters and roles.
2. R2. Organisers create tournaments with a format (single/double elimination, round-robin group stage, Swiss qualifier), rules, schedule and prize information.
3. R3. Registration and check-in windows, seeding (by rating or manual) and waitlists.
4. R4. Automatic generation of groups and brackets, and scheduling of matches into time slots.
5. R5. Match lifecycle: external game servers or organisers report results; the platform validates them, resolves disputes, advances winners and handles no-shows and forfeits.
6. R6. Near-real-time standings, brackets and player/team rating updates.
7. R7. Public, read-heavy surfaces: leaderboards, tournament pages, match pages and live-stream overlay feeds.
8. R8. Notifications for match start, results and schedule changes.
9. R9. Anti-cheat and integrity checks with moderator tooling (bans, result reversals, audit trail).
10. R10. Analytics and reports for organisers (participation, viewership, match statistics).

2.4 Additional context

- C1. Spiky, event-driven load. Traffic is low between events and spikes by orders of magnitude during majors — spectator (read) traffic most of all. The competitive write path must stay responsive while spectator traffic peaks.
- C2. Result ingestion is a continuous stream. Results arrive from many external game servers and must be received, normalised, validated and published with no loss and no double-counting.
- C3. Integrity is critical. Results and rating changes must be tamper-resistant and auditable, and disputes must be traceable; cheating undermines the whole product.
- C4. Global audience. Competitors and spectators are worldwide, so low latency on read surfaces matters.
- C5. Third-party integrations. Game servers, streaming platforms, and payments for entry fees and prizes.
- C6. Independent evolution. Distinct domains (identity, tournaments, result ingestion, ratings, streaming, moderation) are built by different team members and should evolve and deploy independently.
- C7. Reproducible and cloud-ready. The platform must build and run in containers, locally and in the cloud, and scale on demand.
- C8. Availability during events is non-negotiable. An outage during a broadcast final is a reputational disaster.

Forward pointer — the result-ingestion stream. The stream of match results maps naturally onto a producer → transformer → tester → consumer pipeline: game servers produce raw results (C2), a transformer normalises them (server payload → match/player), a tester validates and de-duplicates (C2, C3), and a consumer publishes standings, ratings and overlays (R6). This shapes the reliability and fault-tolerance targets in the architecture-characteristics document.